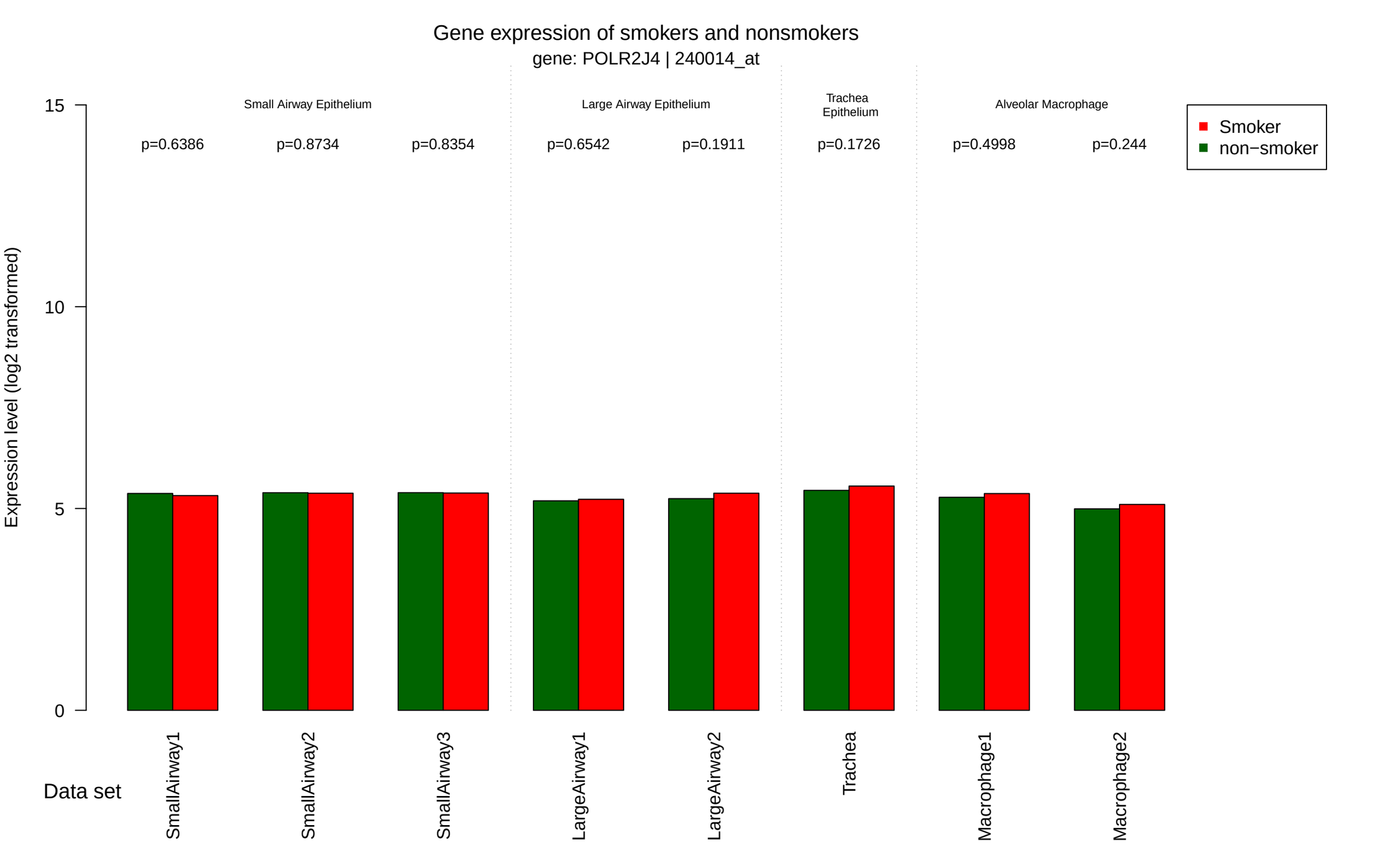
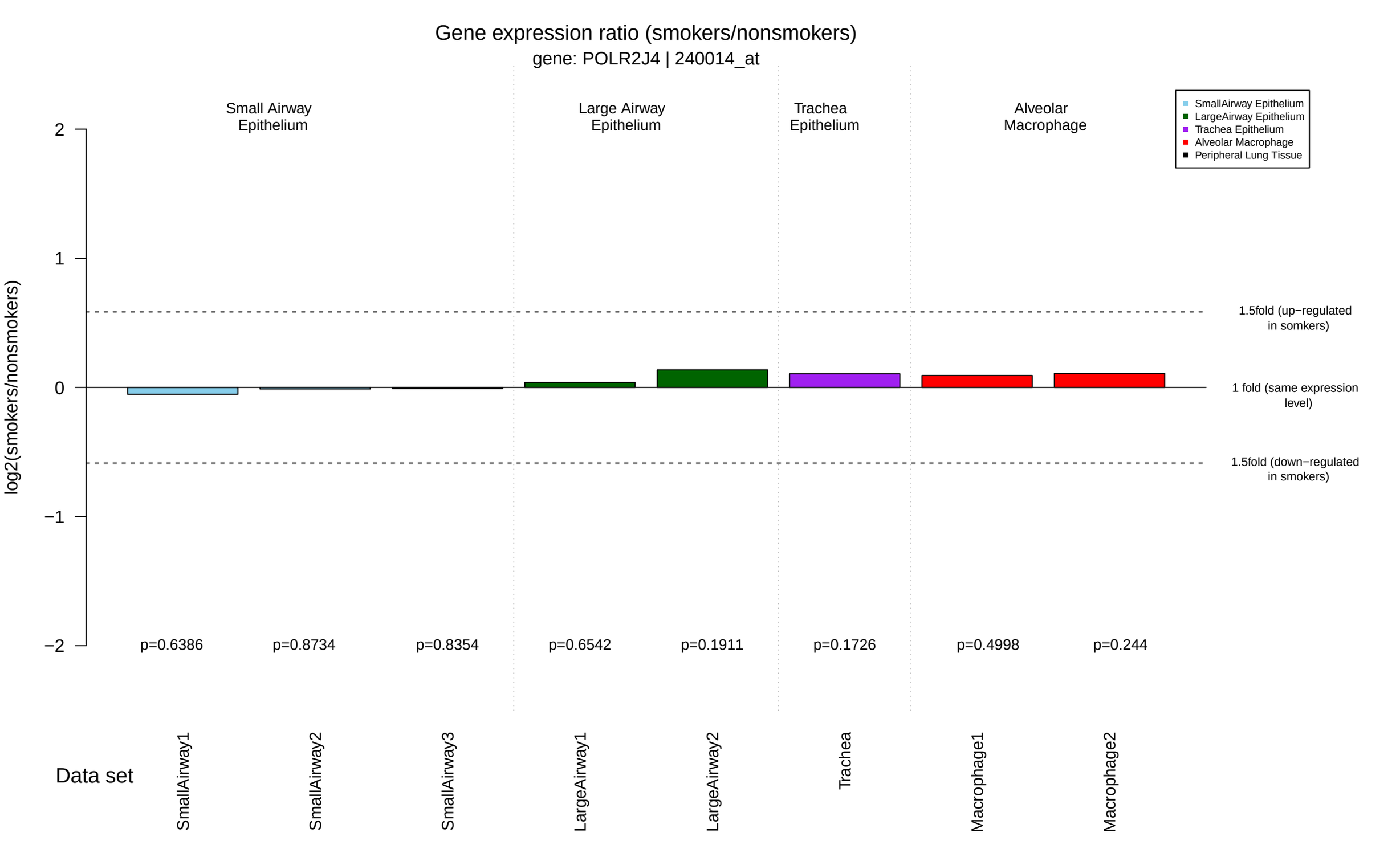
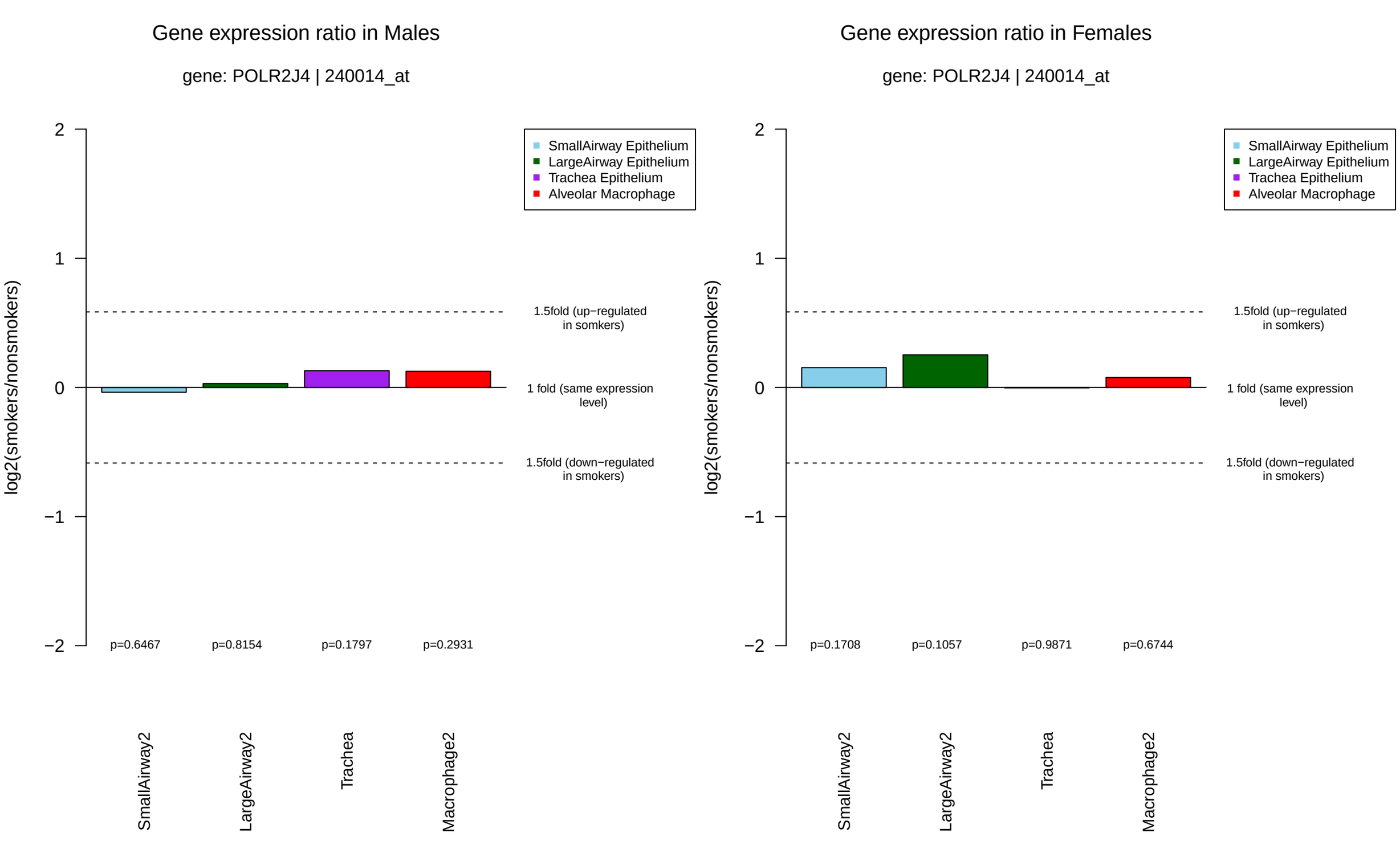
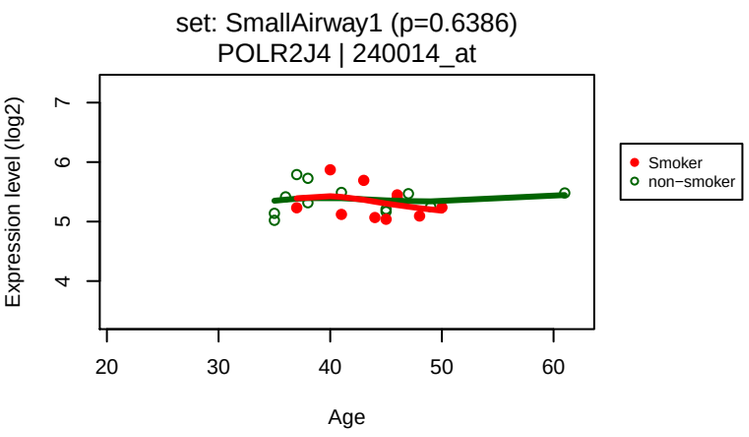
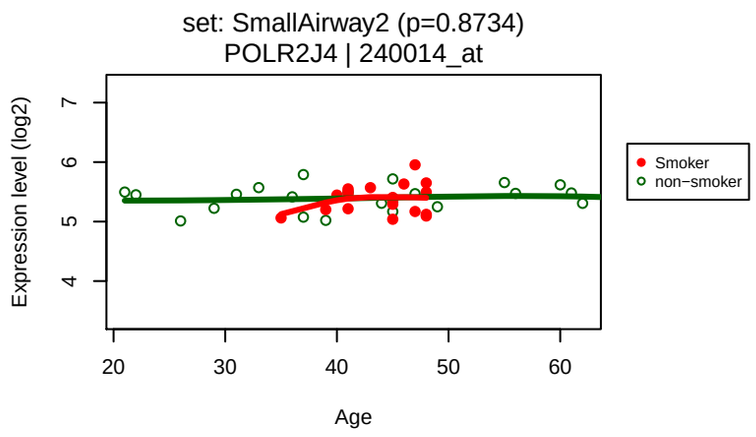
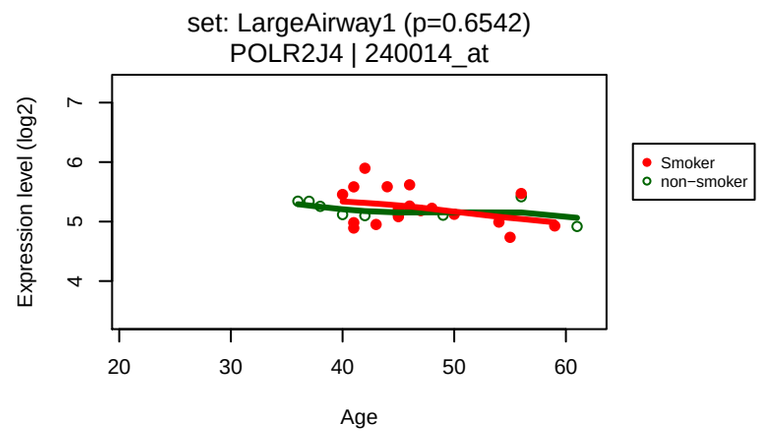

Gene expression of smokers and nonsmokers
gene: POLR2J4 | 240014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6386
p=0.8734
p=0.8354
p=0.6542
p=0.1911
p=0.1726
p=0.4998
p=0.244
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

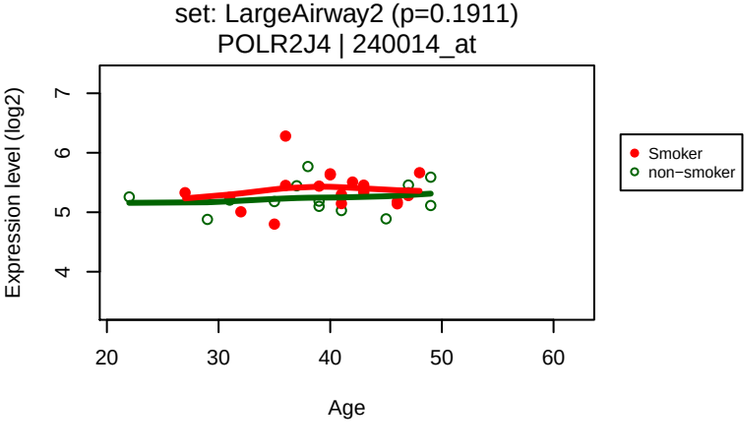

Gene expression ratio (smokers/nonsmokers)
gene: POLR2J4 | 240014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6386
p=0.8734
p=0.8354
p=0.6542
p=0.1911
p=0.1726
p=0.4998
p=0.244
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

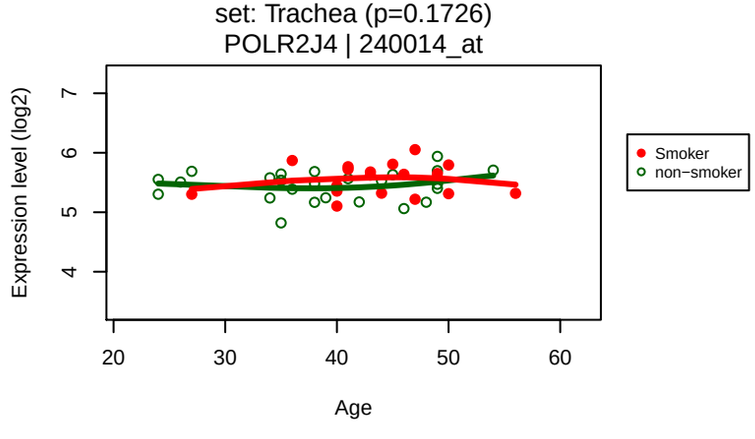

Gene expression ratio in Males
gene: POLR2J4 | 240014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6467
p=0.8154
p=0.1797
p=0.2931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2J4 | 240014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1708
p=0.1057
p=0.9871
p=0.6744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

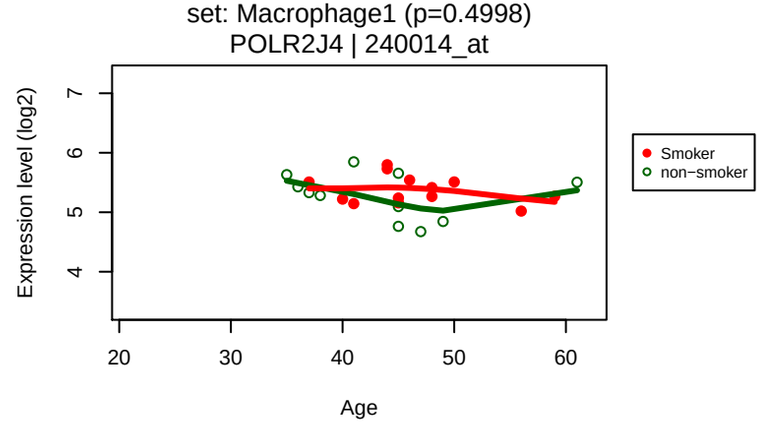

set: SmallAirway1 (p=0.6386)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker

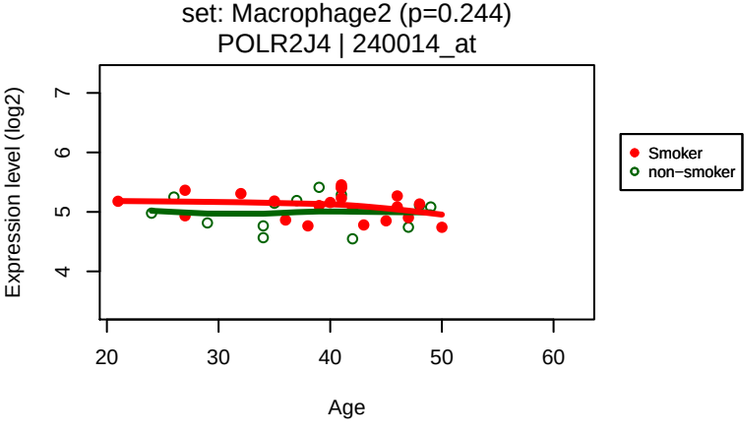

set: SmallAirway2 (p=0.8734)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker

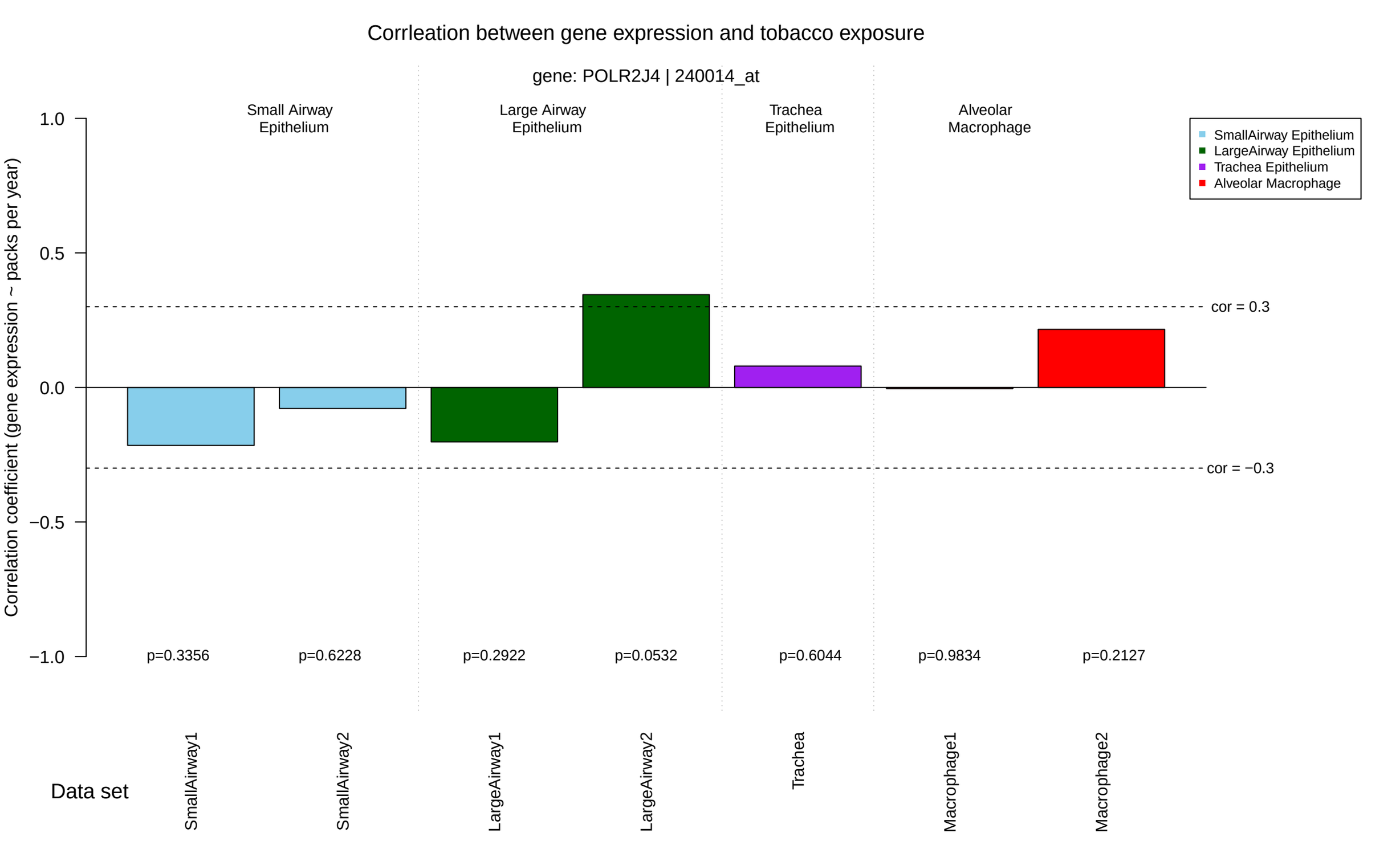

set: LargeAirway1 (p=0.6542)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1911)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1726)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4998)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.244)
POLR2J4 | 240014_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2J4 | 240014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3356
p=0.6228
p=0.2922
p=0.0532
p=0.6044
p=0.9834
p=0.2127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2